

SE-LAB1 USE-CASE FLOW:

NAME: AHIDWISH JAVALI

SRN: PES1UG24CS037

SECTION:5A

GROUP TRAVEL EXPENSE SIMPLIFICATION ENGINE

Use Case ID: UC-06

Use Case Name: Calculate Optimal Settlement

Primary Actors: Group Traveler / Trip Admin

PRECONDITIONS

1. A trip has been created in the system.
2. At least two travelers are associated with the trip.
3. Expenses have been recorded for the trip.
4. Each recorded expense has a valid payer and expense allocation.
5. Required currency conversions are available for foreign-currency expenses.

POSTCONDITIONS

1. The net balance of every traveler has been calculated.
2. Debtors and creditors have been identified.
3. An optimal debt settlement graph has been generated.
4. The total group balance is zero.
5. Minimum required peer-to-peer payment transactions are displayed.

MAIN SUCCESS SCENARIO

1. The Group Traveler or Trip Admin selects Calculate Optimal Settlement.
2. The system retrieves all recorded expenses and participant shares.
3. The system retrieves applicable daily exchange rates.
4. The system converts foreign-currency expenses into the trip's base currency.
5. The system calculates each traveler's net balance.
6. The system identifies debtors and creditors.

7. The system applies the min-flow settlement algorithm.
8. The system generates the optimal settlement graph.
9. The system verifies that the resulting balances are net zero.
10. The system displays the recommended payment transactions.
11. The use case ends successfully.

ALTERNATE FLOW A1: Exchange Rate Unavailable

- A1. During step 3, the required daily exchange rate cannot be retrieved.
- A2. The system identifies the affected foreign-currency expense.
- A3. The system displays an error indicating that settlement cannot continue.
- A4. The system does not generate the settlement graph.
- A5. The user is instructed to retry once the exchange rate becomes available.